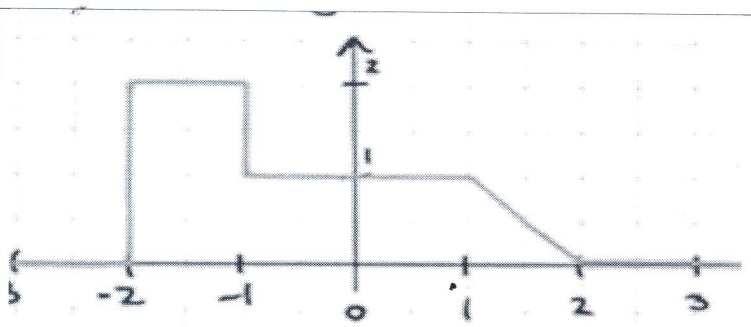
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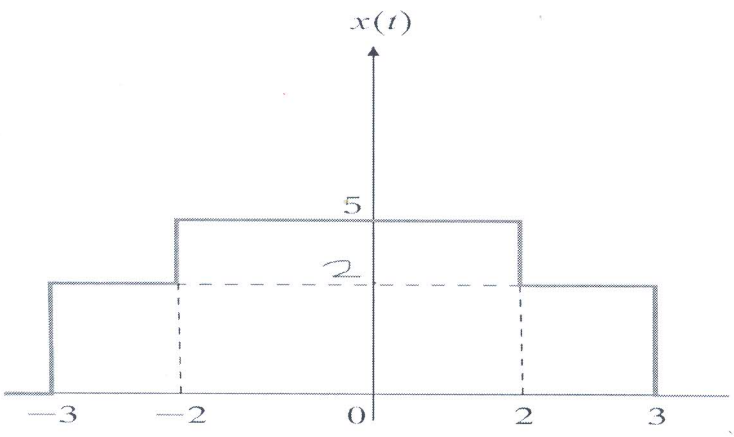
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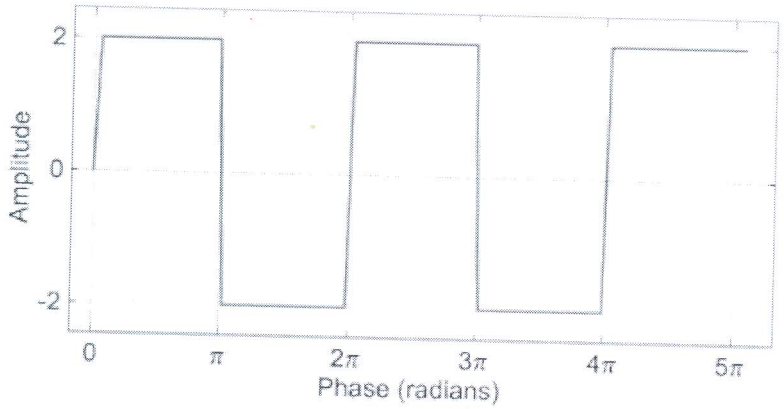
Programme: B. TECH (CSE - DS)/B. Tech (DS)/B. Tech Intg. (DS)/MBA Tech (DS)
Year: II / IV/Sem 4 Re-exam/Sem 8 Re-exam(Exam Year: 2025-2026)
Subject: Introduction to Data, Signal and Image Analysis
Date: 11 Jul 2026 **Time: 02:00 pm to 05:00 pm (03:00 Hrs.)**
Max. Marks: 100

RE EXAMINATION (Acad. Year 2025-2026)

Instructions:
1. This question paper contains 3 pages.
2. Question No. _1___ is compulsory.
3. Out of remaining questions, attempt any __4___ questions
4. In all ___5_ questions to be attempted.
5. All questions carry equal marks
6. Answer to each new question to be started on a fresh page.
7. Figures in brackets on the right hand side indicate full marks
8. Assume Suitable data if necessary

Q1		Answer briefly:	[20]
CO1 **BL1**	a.	Write Laplace Transform for:- Sin5t, Cos7t, e^{6t}, e^{-2t}, 10	[5]
CO1 BL3	b.	Solve to find Z- Transform for :- $X(n) = 10, 12, 15$	[5]
CO2 BL1	c.	List the advantages of Butterworth Filter Design	[5]
CO1 BL2	d.	Compare Analog vs. Digital signal	[5]
Q2 CO3 **BL3**	A	Explain the significance of sampling to convert continuous time into discrete. Apply sampling concept on a continuous time signal $x(t) = 50 \sin 125\pi t$ and find its discrete signal.	[10]
Q2 CO1 **BL3**	B	 Consider the signal x(t) given in the figure. Perform and sketch x (-t), X (t+1), X (-t-1), X(5t)	[10]

Q3 CO3 **BL3**	A	Determine Y(t) by using circular convolution. Given – $x(t) = 7,8,9,10$ *and* $h(t) = 1,2,-1,3$	[10]
Q3 CO1 **BL4**	B	Evaluate Fourier Transform for the following signal: -  	[10]
Q4 CO1 **BL3**	A	Construct and sketch the signal X(t) = 5u(t+3) – u(t+2) – r(t+1) +r(t-2)	[10]
Q4 CO1 **BL4**	B	Evaluate- $L[\int_0^t e^{-15t}.t.sin20t\ dt]$	[10]
Q5 CO4 **BL2**	A	Compare Impulse Invariant vs. Bilinear Transformation	[10]
Q5 CO4 **BL2**	B	Write a detailed note on image compression with respect to:- Goal, Need, Types of Redundancy in Images, Lossless Compression, Lossy Compression	[10]
Q6 CO2 **BL4**	A	Compare Finite Impulse Response and Infinite Impulse Response filters in terms of the following aspects: 1. Quantization Effects 2. Numerical stability 3. Flexibility in design 4. Implementation cost 5. Pole-zero structure	[10]
Q6 CO4 **BL3**	B	A grayscale image has **4 intensity levels (0–3)** with the following histogram: Intensity level: 0, 1, 2, 3 Number of pixels: 1, 1, 2, 4 **Perform histogram equalization** and determine the **new intensity values**.	[10]

Q7 CO4 **BL2**	A	Write a detailed note on Mean Filter focusing on its: - Definition, Mathematical representation, Kernel, working principle, example.	[10]
Q7 CO1 **BL3**	B	Apply Fourier series (trigonometric form) for the given signal  Points on X-axis:- 0, π, 2π, 3π, 4π, 5π… , Points on Y-axis:- +2, -2, Signal starts at zero.	[10]